

Practice Problems

1. A bag contains 5 red balls and 3 blue balls. What is the probability of drawing 2 red balls in a row without replacement?
2. If $X \sim \text{Binomial}(20, 0.3)$, what is $E[X]$ and $\text{Var}(X)$?
3. Customer arrivals follow $\text{Exponential}(\lambda = 3 \text{ per hour})$. What is the probability the next customer arrives within 15 minutes?
4. Two independent investments have $E[X_1] = 8\%$, $\text{Var}(X_1) = 9$ and $E[X_2] = 6\%$, $\text{Var}(X_2) = 4$. If you invest equally in both, what is $E[\text{Portfolio}]$ and $\text{Var}(\text{Portfolio})$?
5. In a population, 2% have a gene. A test detects it with 98% accuracy (both ways). If you test positive, what is the probability you have the gene?